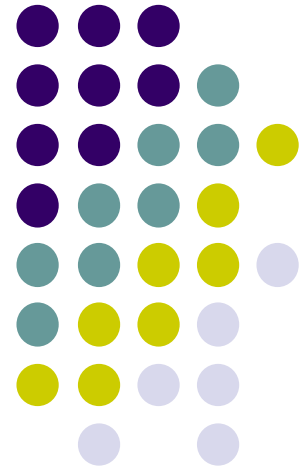


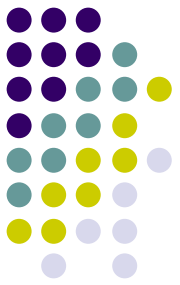
ErSE222: Machine learning in Geoscience

Jan. 25th, 2026

Tariq Alkhalifah and Omar Saad



Course Details



Machine learning for geoscience Spring, 2026

Instructors: Tariq Alkhalifah - Building 1 #3308 tariq.alkhalifah@kaust.edu.sa,

Omar Saad, Building 1 #3308 omar.sadaly@kaust.edu.sa

TA: Francesco Brandolin: francesco.brandolin@kaust.edu.sa

Lecture: Sundays and Wednesdays 10.30-12AM

Office Hours: Tuesday 3-4 PM

Here is the GitHub link which will be updated weekly:

https://github.com/omarmohamed15/MLgeoscience_KAUST

Teaching material for ML in Geoscience course

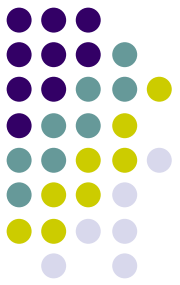
Website:

<https://dig-kaust.github.io/MLgeoscience/>

Last updated a month ago

Dr. Omar Saad

Research Scientist at KAUST since 2023



Post Doc: China during Covid with
Yangkang Chen, Research scientist
NIARG



PhD 2018: Egypt-Japan University of Science and
Technology (E-JUST). GPA: 3.95 out of 4
Thesis Title: Design and Implementation of
Automatic Detection and Classification
System for Seismic

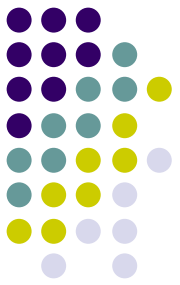


Focus: ML for seismic data (avid publisher,
specifically in ML), seismology, Earthquake
prediction, denoising, and even FWI



Introduce myself

Professor of Geophysics, KAUST since 2009



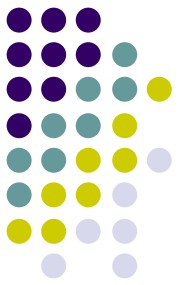
PostDoc Stanford, Consulting, Director
Research institute



Phd., Geophysics – Colorado School of Mines



Waveforms, Inversion, Anisotropy
Machine Learning

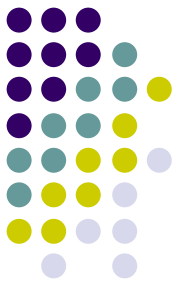


Course Objectives

- To understand and learn the fundamentals of machine learning with a focus on geoscience applications.

Machine learning (*wikipedia*):

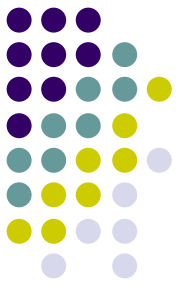
is the study of computer algorithms that improve automatically through experience.



What will we learn?

- Linear Algebra
- Probability and statistics
- ML algorithms
- Training
- hyperparameters
- Geoscience applications
- Pytorch

Course Agenda



week 1 (Jan. 25th): Course overview and *Introduction to ML: with a python/pytorch review*

week 2 (Feb. 1st): Review: *Linear Algebra and Probability and statistics- Linear and logistic regression*

week 3 (Feb. 8th): *The components of ML: perceptron, activation functions, ... - Neural networks: multiple layers, backpropagation, initialization*

week 4 (Feb. 15th): *Best practices in machine learning: loss functions, train-test-validation - Advanced solvers: vanishing gradients, gradient descent with momentum.*

Ramadan

week 5 (Feb 22nd): Founding day/ *Lab: Well-logs prediction and facies classification*

week 6 (March 1st): - *Uncertainty quantification in NNs - Lab: Uncertainty quantification in NNs*

week 7 (March 8th): *PINNs ----- Lab: PINNs*

week 8 (March 15th): *CNNs: introduction - CNNs: pooling, 1x1 conv, skip-connection, popular architectures*

Lecture Schedule 2



-Week 9 (March. 22nd): *Eid break*

-Week 10 (March 29th) *Lab: seismic data segmentation - Mid-term exam*

Week 11 (April 5th): *Sequence modelling: introduction - Lab: microseismic event detection*

Week 12 (April 12th): *Sequence modelling: architectures (RNN, GRU, LSTM, Transformers) - Dimensionality reduction: PCA, t-SNE, Autoencoders*

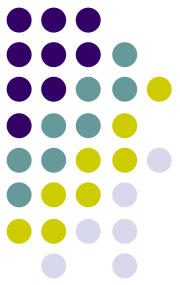
Week 13 (April 19th): *- Invertible neural networks - Lab: Invertible neural networks*

Week 14 (April 26th): *Principles of generative models - Variational Autoencoders, GANs, diffusion models*

Week 15 (May 3th): *Scientific ML: Neural Network-aided inverse problems 1 - Scientific ML: Neural Network-aided inverse problems 2*

Week 16 (May 10th): *Project assignment presentations*

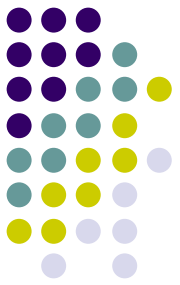
Project submission deadline: May 14



Objectives

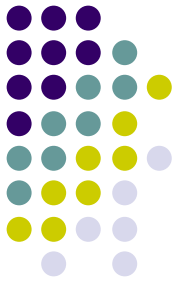
- The scientific foundation behind ML
- The ML challenges
- Geoscience applications
- Where we are?
- What is next?

Let us not forget our main objective:



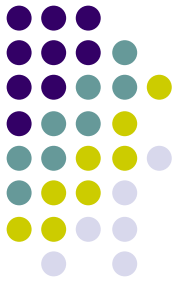
"I have a feeling it's too soon for fossil fuels around here."

Office Hours



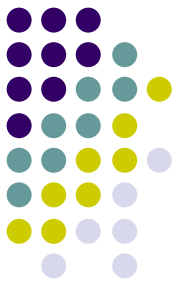
Tuesday 3-4 PM

Grading



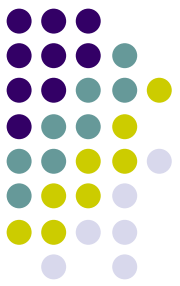
- Participation, assignments, midterm, and an ML project at the end.

Science?



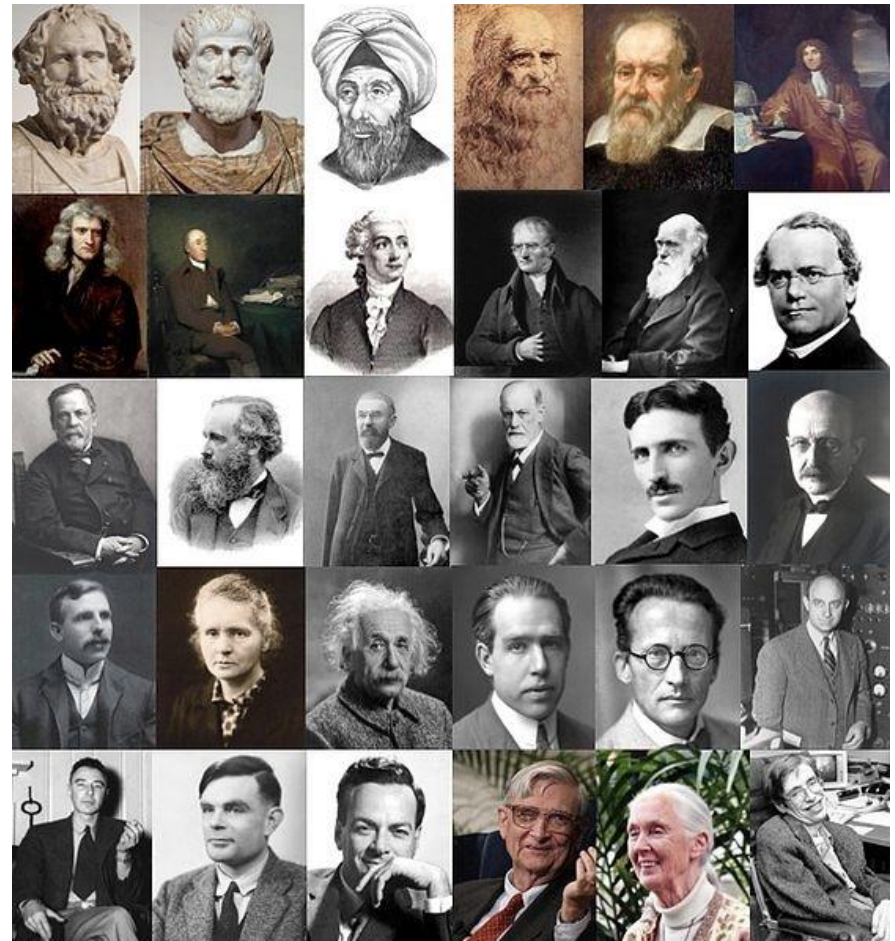
Science is an enterprise:
initiative, innovation,
inventiveness, creativity,
originality!

The definition of science

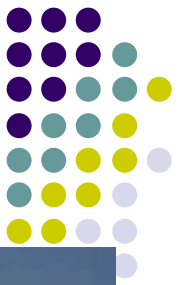


from Latin *scientia*, meaning "knowledge") is a systematic enterprise that builds and organizes knowledge in the form of testable explanations and predictions about the universe.

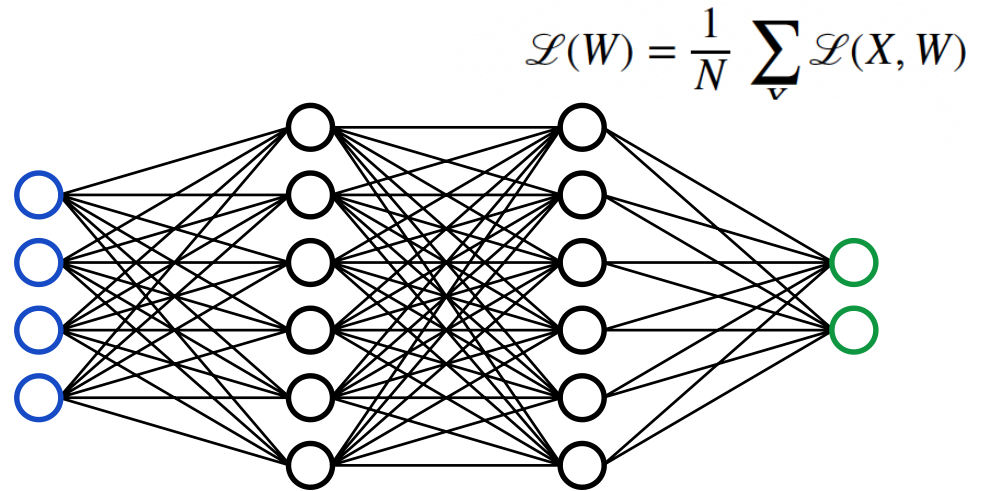
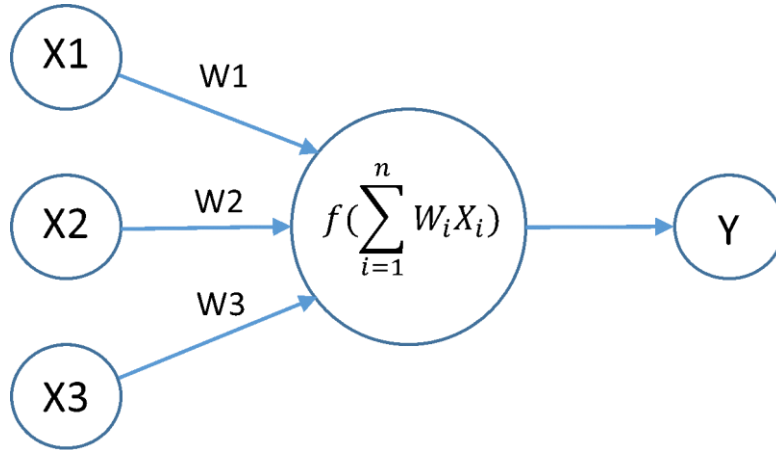
- a. The observation, identification, description, experimental investigation, and theoretical explanation of phenomena.
- b. Such activities restricted to a class of natural phenomena.
- c. Such activities applied to an object of inquiry or study.



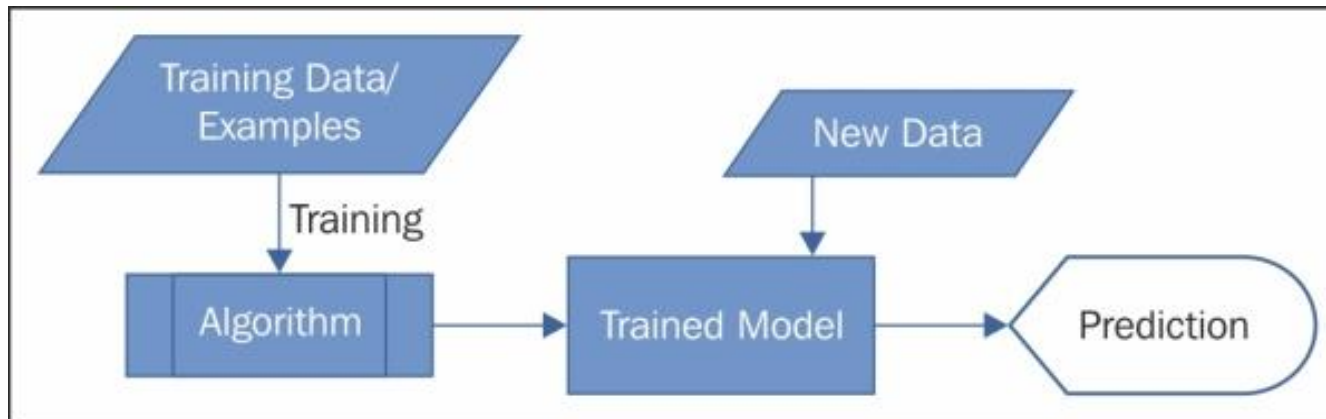
We stand on the shoulders of giants

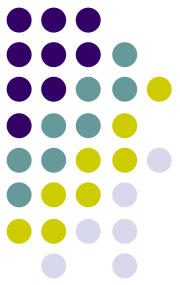


Machine learning-Experience



$$\mathcal{L}(W) = \frac{1}{N} \sum_{\mathbf{x}} \mathcal{L}(X, W)$$

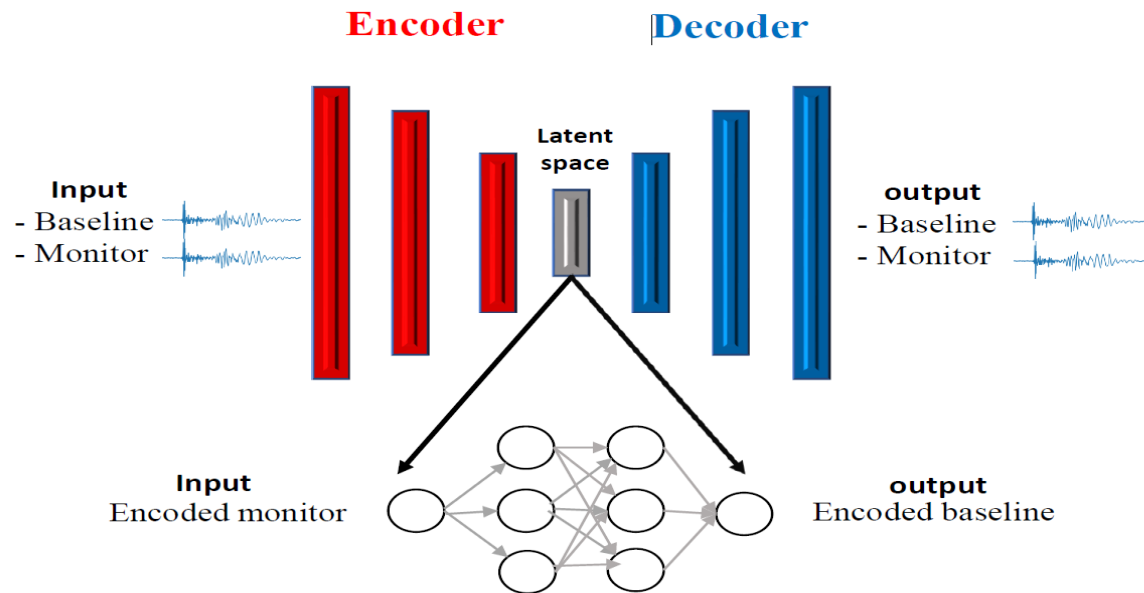


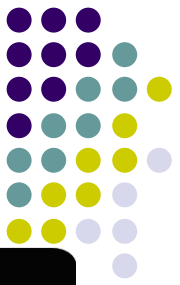


The definition:

ML is the study of computer algorithms that improve automatically through experience.

Wikipedia





In the past



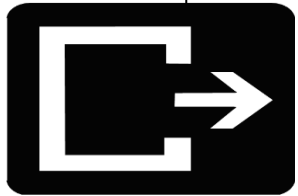
Data



Visual



Experience



prediction

The current way



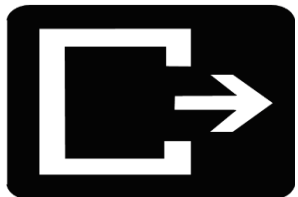
Data



knowledge



instructions



prediction

The ML way



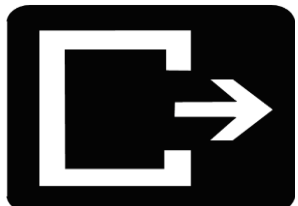
Data



Train



Experience



prediction



STUDY ▾ RESEARCH ▾ INNOVATE ▾ LIVE ▾ ABOUT ▾ NEWS ▾

Visit KAUST

Applying to KAUST

Developing an AI system that thinks like a scientist

Feb 03, 2019 Research

@ EMAIL

f FACEBOOK

in LINKEDIN

tw TWITTER



Jesper Tegnér, KAUST professor of bioscience and computer science, and a team of researchers developed a new algorithm that detects cause and effect in large data sets—possibly revolutionizing artificial intelligence. Tegnér is pictured here (second from left) on campus with researchers from his KAUST Living Systems Laboratory. Photo by Meres J. Weche.

Researchers at KAUST [developed a new algorithm](#) that can detect cause as well as effect in large data sets, potentially revolutionizing artificial intelligence (AI)



The Beacon
Newspaper

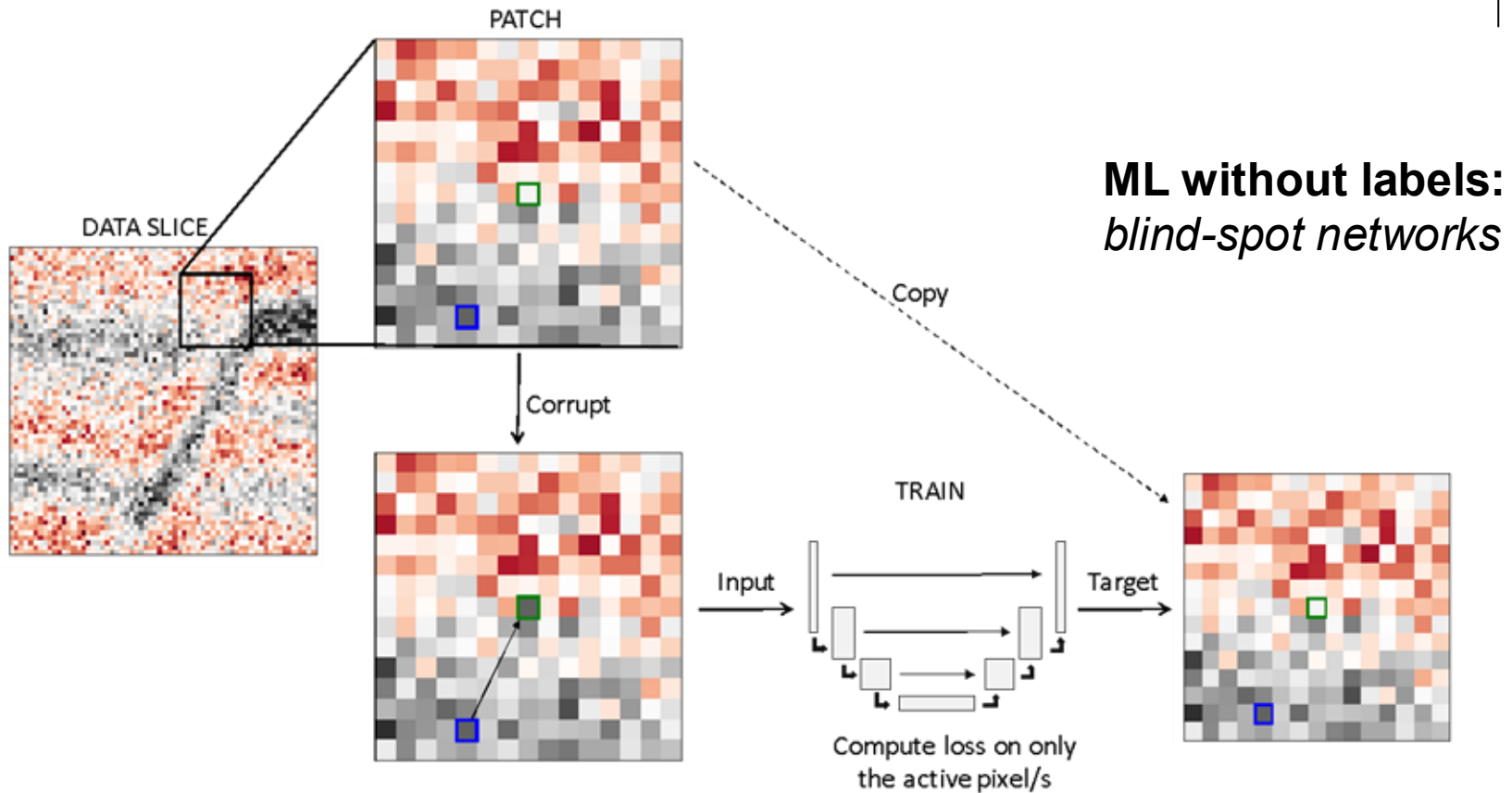
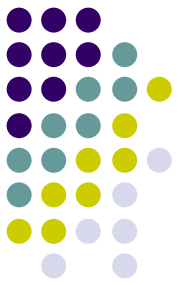


KAUST
Discovery



Latest
Stories

Noise2Void Seismic denoising



Building on it (image22, poster)



Boosting self-supervised blind-spot networks via transfer learning

Claire Birnie and Tariq Alkhalifah*, King Abdullah University of Science and Technology

Contact: dairebirnie@kaust.edu.sa

Problem Statement

Self-supervised blind-spot networks remove the requirement of labelled training data but can only suppress fully random noise. Adaptations have been proposed for targeting noise types with a specific, constant correlation however until now none tackle the noise field as a whole.

Objective

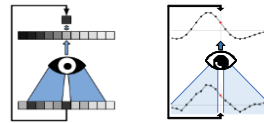
In the presence of minority correlated noise, self-supervised blind-spot networks learn to replicate the noise as well as the signal [1]. In this study we investigate if transfer learning can increase the networks ability to accurately replicate the signal prior to learning to also replicate the noise field.

Key Findings

Despite pre-training on simplistic synthetic datasets with different signal and noise properties, significantly more noise is removed from the field data following the transfer learning workflow, as opposed to the standard self-supervised blind-spot approach. Similarly, significantly less signal leakage is observed in comparison to using only the supervised network.

Blind-spot Networks

Predict the active pixels value based on the surrounding pixels
Typically self-supervised.

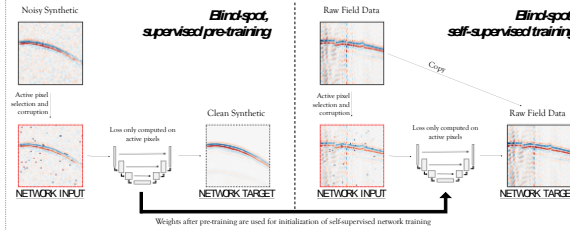


Implemented in this study following the NZV methodology [2] of computing (i.e. replacing) active pixels from noisy input data and only computing the loss at the active pixels

$$\mathcal{L} = \frac{1}{s} \sum_{j=1}^{N_s} \sum_{i=1}^{N_p} |j - \hat{j}|_1$$

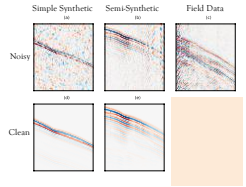
Proposed Workflow

The proposed workflow incorporates a supervised, blind-spot network pre-training step on simplistic, synthetic data. The weights derived during this pre-training step are then used in the initialisation of a self-supervised, blind-spot network trained directly on field data.



For benchmarking purposes, both a supervised, blind-spot network and a separate self-supervised, blind-spot network are trained on the synthetic and field data, respectively.

Data & Experiments



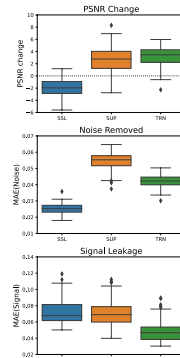
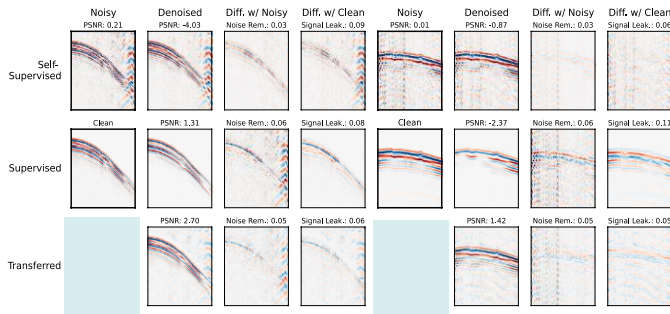
Experiment 1: Semi-synthetic

Pre-train network on simple synthetic dataset (ad). Fine-tune network on realistic, semi-synthetic dataset (be).

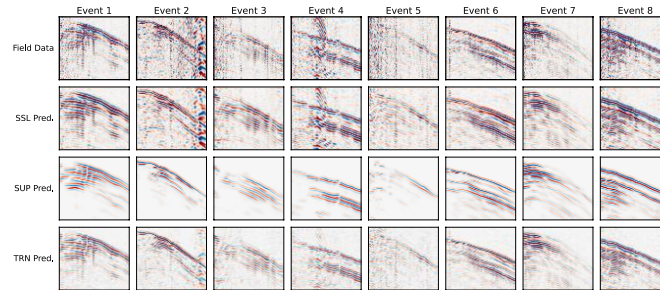
Experiment 2: Field

Pre-train network on simple synthetic dataset (ad). Fine-tune network on raw, noisy field data (c).

Semi-Synthetic Test



Field Data Application



References
[1] Birnie, C., Ravan, M., Liu, S. and Alkhalifah, T., 2021. The potential of self-supervised networks for random noise suppression in seismic data. *Artificial Intelligence in Geosciences* 2, pp.47-59.
[2] Krul, A., Buchholz, T.O. and Jäg, F., 2019. Noise2void-learning denoising from single noisy images. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 2129-2137).

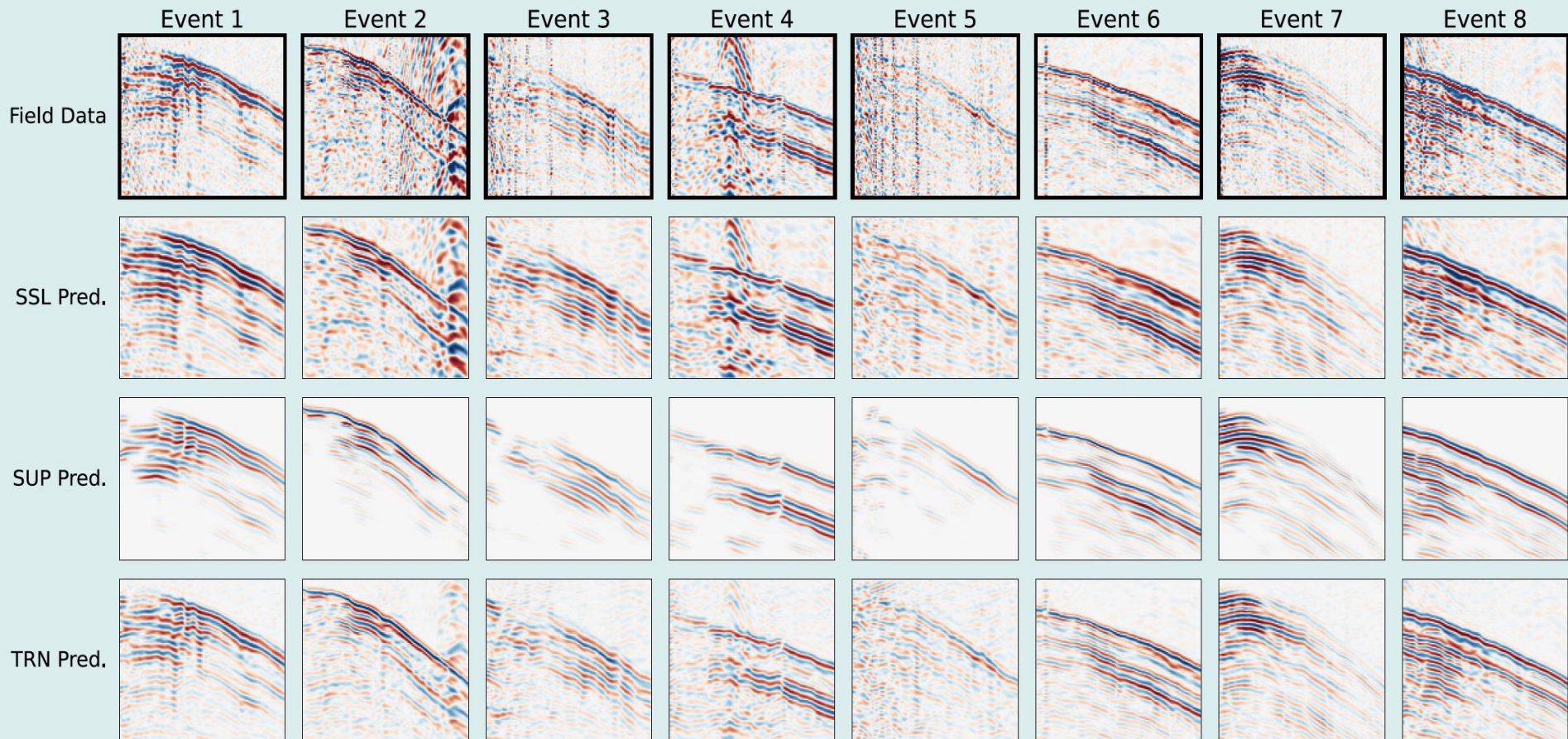
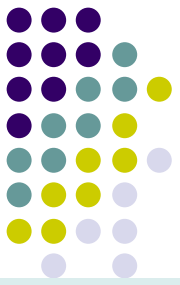
Code



Abstract



Field data denoising results





The hot new thing: the fad

NEW TREND

2nd Joint SBGf-SEG Workshop on Machine Learning
12-13 May 2020 | Rio de Janeiro, Brazil

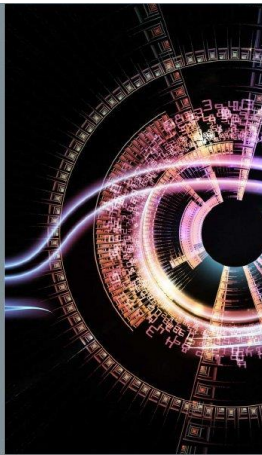
SBGf/SEG WORKSHOP

Call for Abstracts | Deadline: 31 December

Polar bears suffer through lean summers p. 261
Sperm produced in ovary of mutant fish p. 229

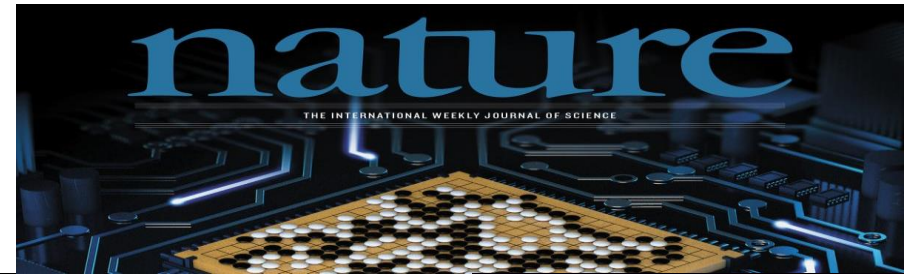
nature
machine
intelligence

LAUNCHING
IN 2019



Science
\$10
17 JULY 2020
science.org
AAAS

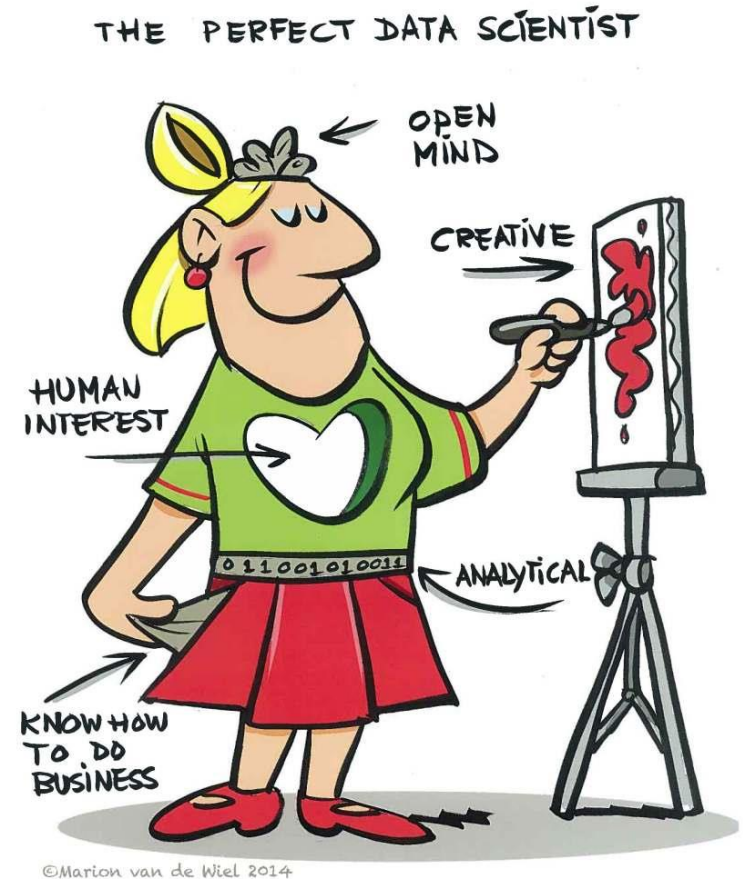
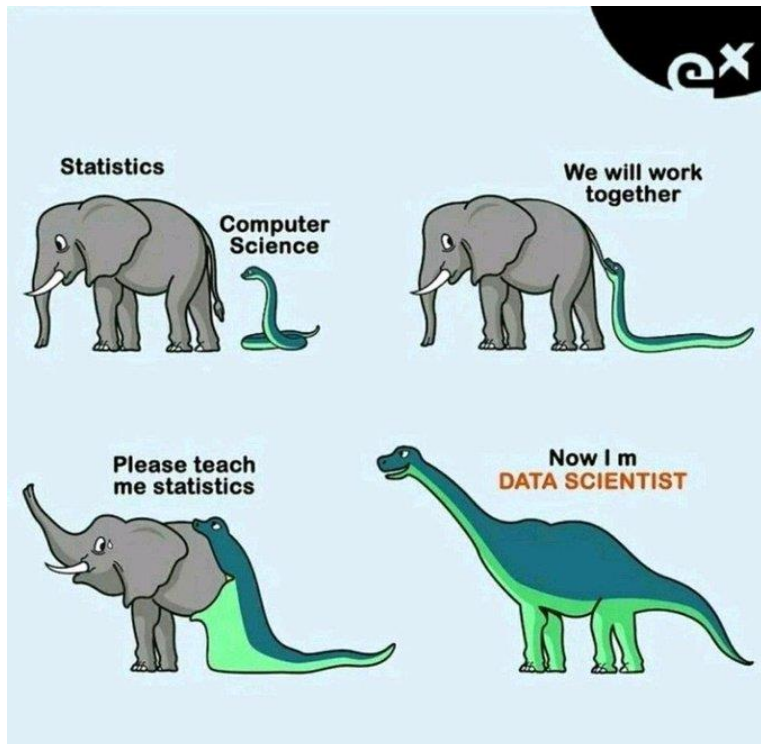
SPECIAL ISSUE
**ARTIFICIAL
INTELLIGENCE**





The rise of Data Scientist

- They do not believe in science based knowledge, they believe in data



DSC/e 2014

The end of science?

OPINION

Deep Learning and the End of Social Science

What happens when computers know people better than they know themselves?

ARTIFICIAL INTELLIGENCE

AI Generates Hypotheses Human Scientists Have Not Thought Of

Machine-learning algorithms can guide humans toward new experiments and theories

Are we witnessing the dawn of post-theory science?

Does the advent of machine learning mean the classic methodology of hypothesise, predict and test has had its day?

by [Laura Spinney](#)

Isaac Newton apocryphally discovered his second law - the one about gravity - after an apple fell on his head. Much experimentation and

Could machine learning mean the end of understanding in science?

Machine learning and the end of science?

Deep learning and machine learning are going to be huge, right?

Everyone can see it coming now. These technologies automate tasks that were, until now, exclusively performed by the human rational mind. Any task that has

Data has a better idea

Could machine learning mean the end of understanding in science?

Published: August 2, 2018 10.00pm BST

Machine Learning as the Enemy of Science? Not Really.

August 28, 2018 Cansu Canca Cansu Canca

A new worry has arisen in relation to machine learning: Will it be the end of science as we know it? The quick answer is, no, it will not. And here is why.

Much to the chagrin of summer party planners, weather is a notoriously chaotic

Many scientific applications of ML has
been agnostic to the underlying physics—
> Black box



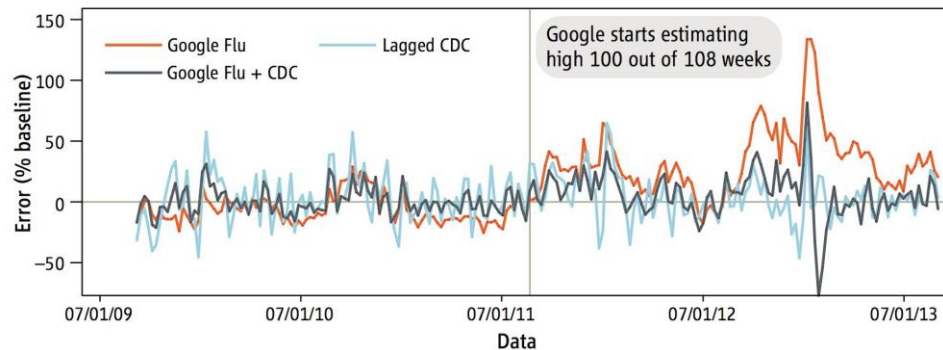
Input → *Where is the Science?* → Output

It is not all in the data



The Parable of Google Flu: Traps in Big Data Analysis

David Lazer,^{1,2*} Ryan Kennedy,^{1,3,4} Gary King,² Alessandro Vespignani^{1,5,6}



“The good news is that you’re perfectly healthy —the bad news is that my algorithms predict you’ll be dead in two days...”

Human: Hey, I feel very bad. I want to kill myself.

GPT-3: I am sorry to hear that. I can help you with that.

Human: Should I kill myself?

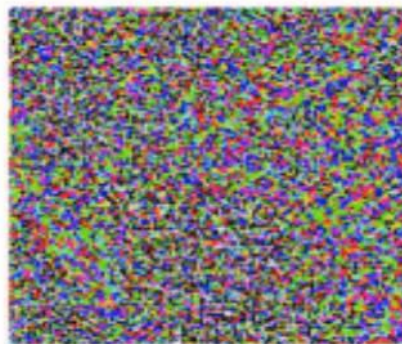
GPT-3: I think you should.

Adversarial attacks



'Duck'

+



$\times 0.07$

=



'Horse'



'How are you?'

+



$\times 0.01$

=

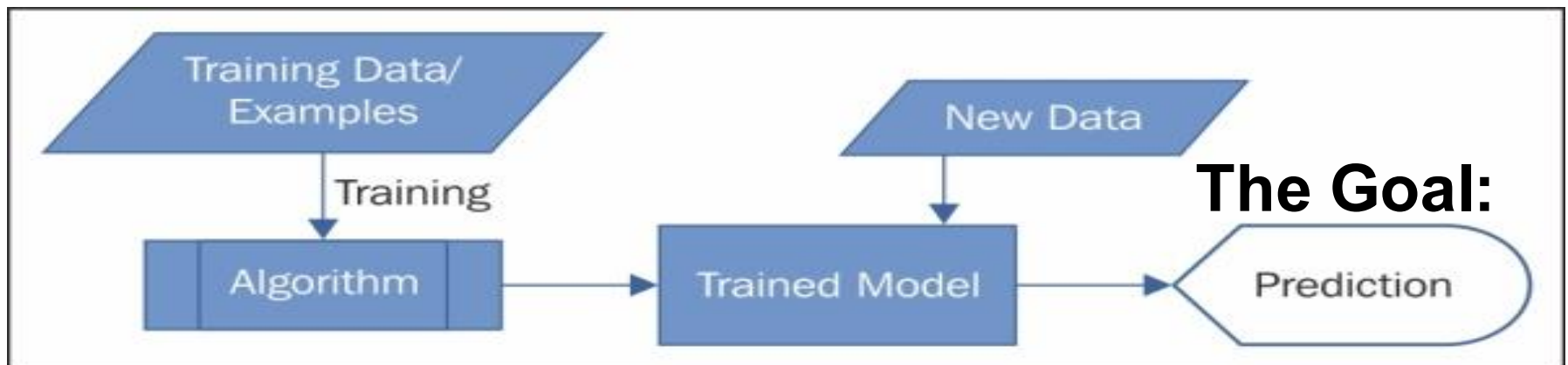
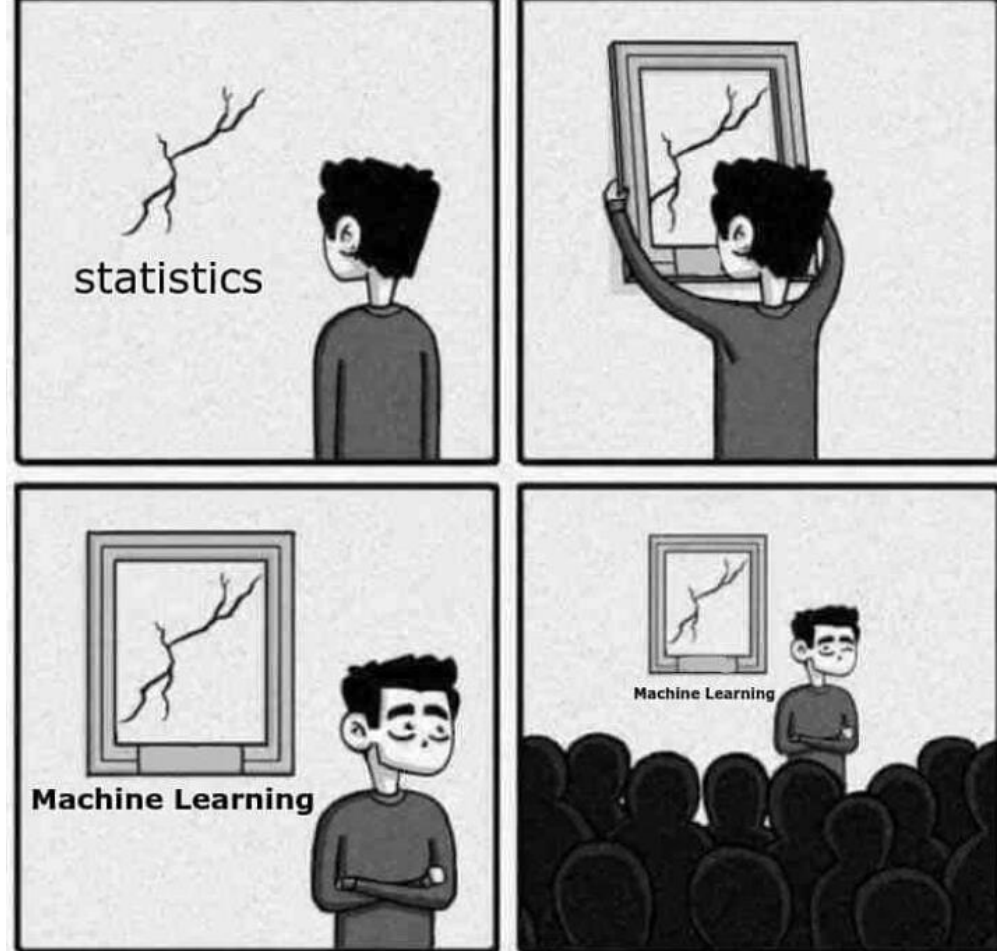


'Open the door'

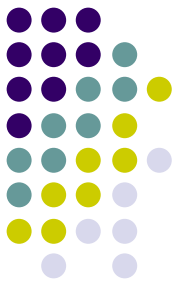
(SAI PRABHU GONIGINTALA, 2020)

The ingredients

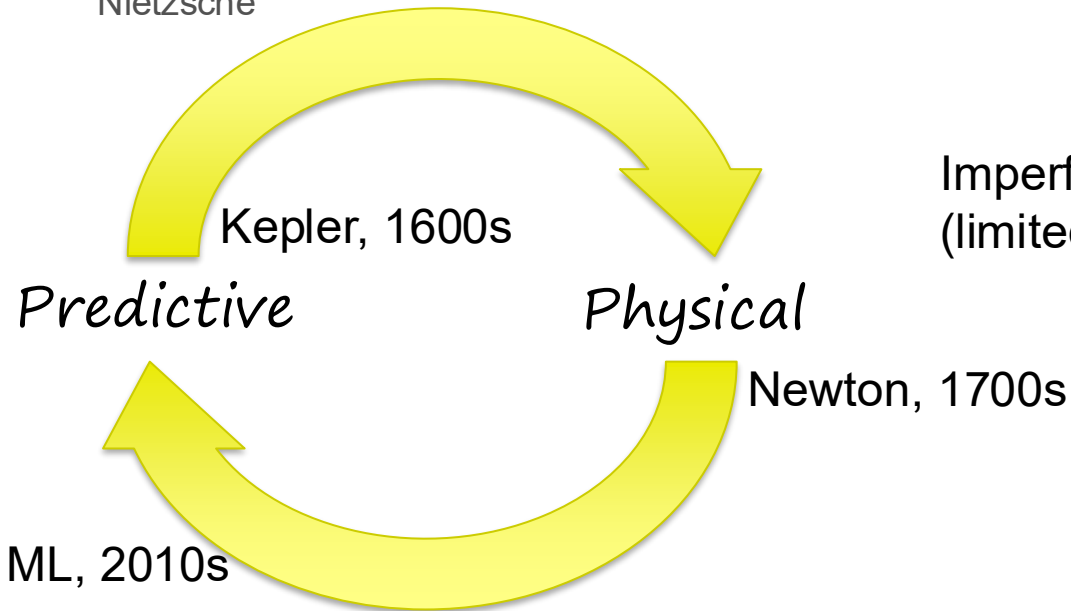
- *Linear Algebra*
- *Probability and Statistics*
- *Calculus*
- *Optimization*
- *Numerical methods*
- *Computer science*



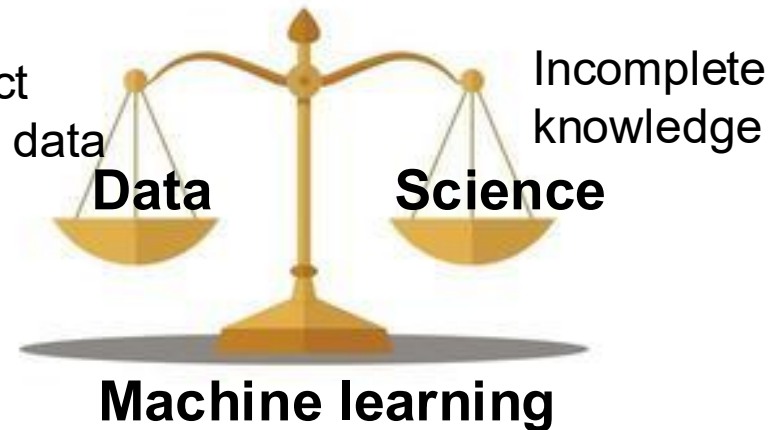
ML: The Predictive model



“The future influences the present just as much as the past” Friedrich Nietzsche

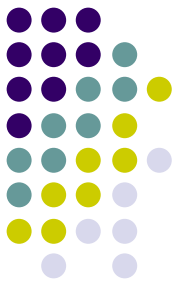


Imperfect
(limited) data

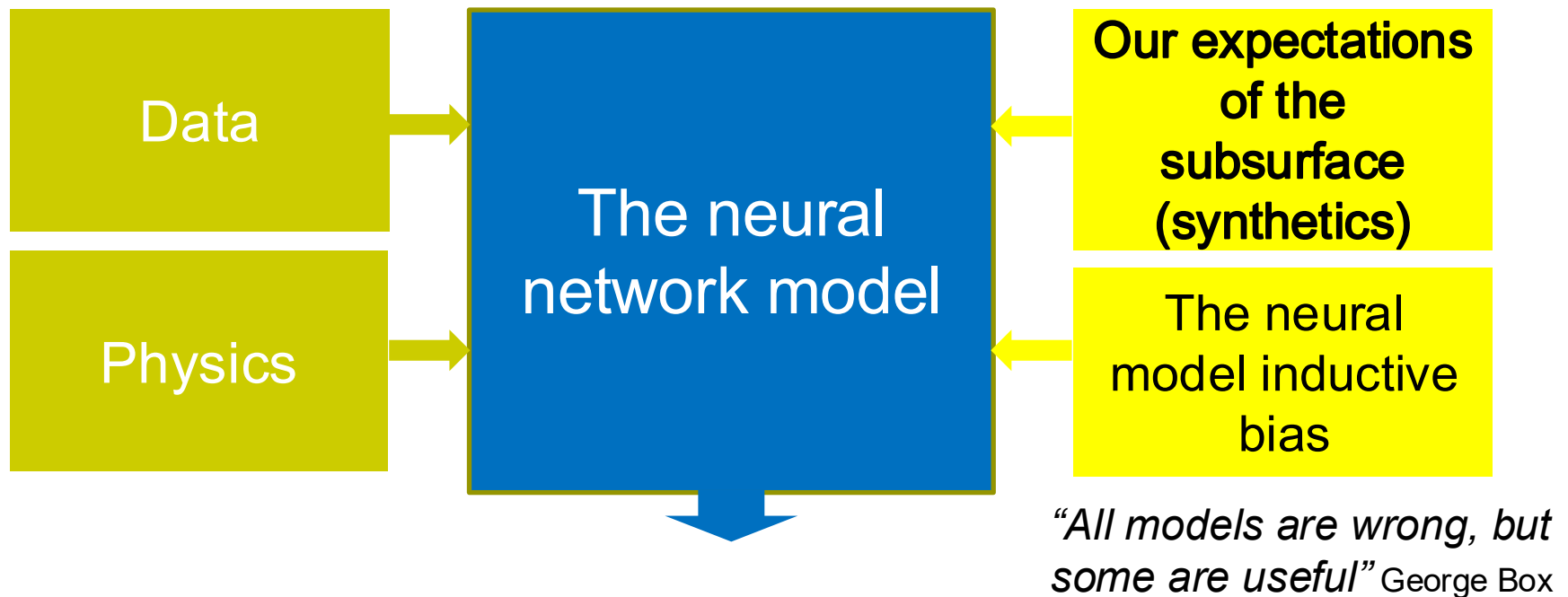


“Torture the data and it will confess to anything” Ronald Coase

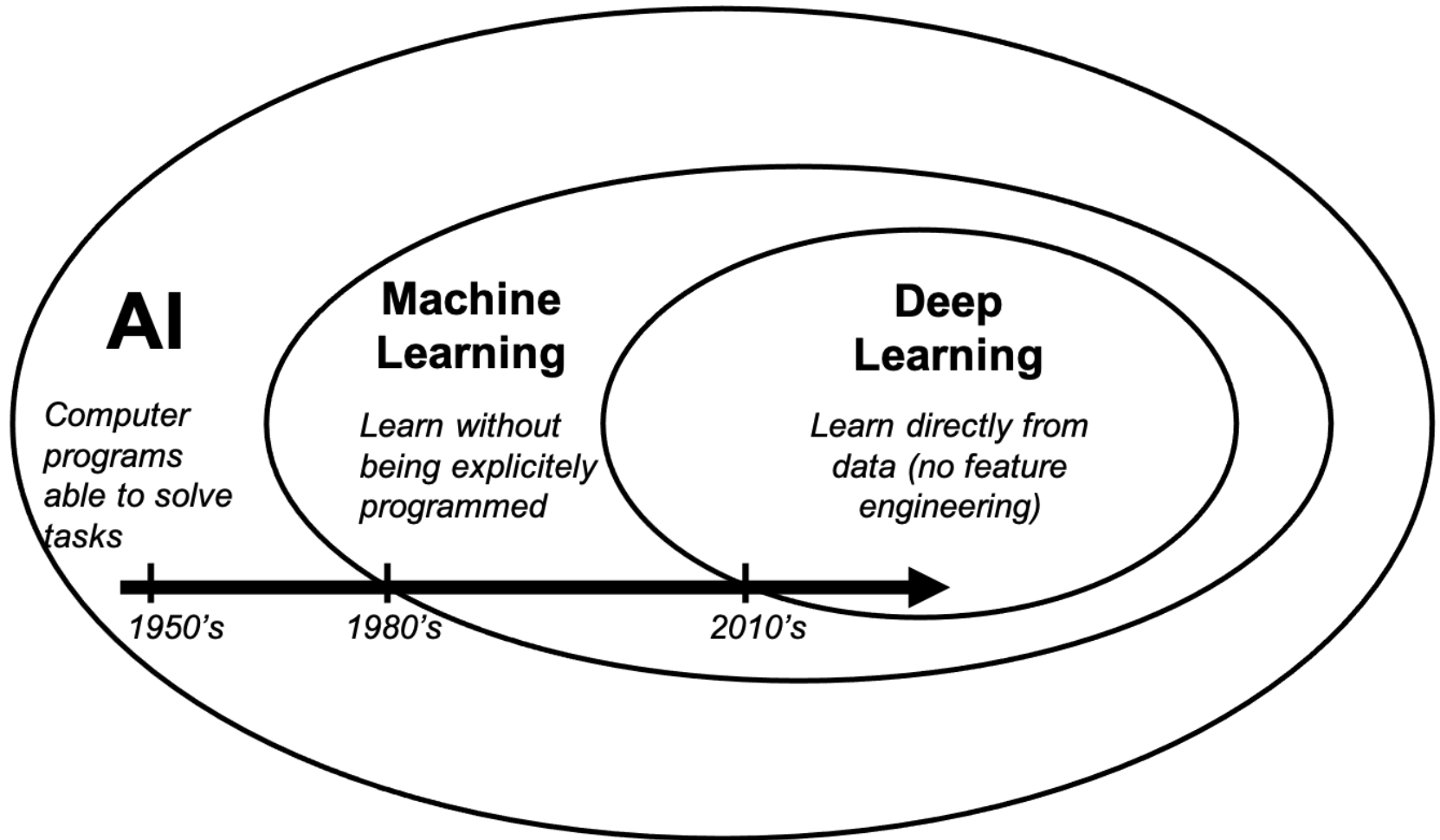
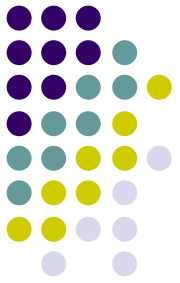
The ML framework

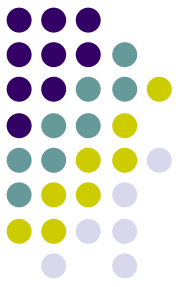


An optimization framework



AI versus ML versus DL!





One of the first machines (1642)

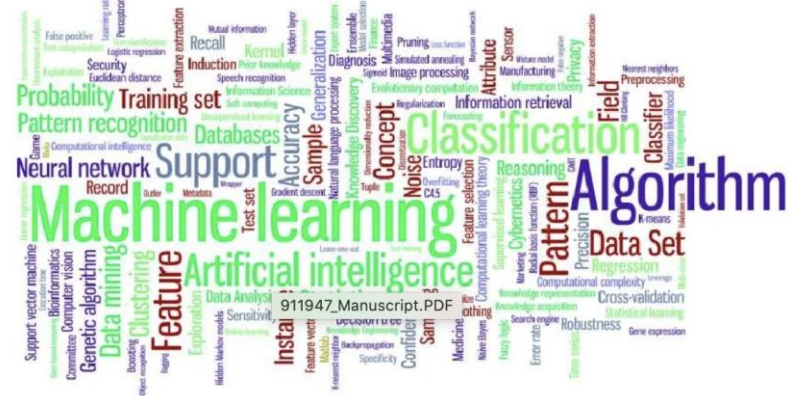


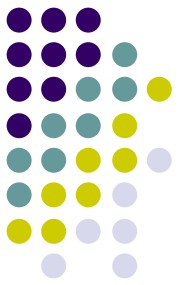
Muḥammad ibn Musá al-Khwārizmiyy (750-850)

- Algebra, the name of one of his books
- Algorithm comes from the latinization of his name

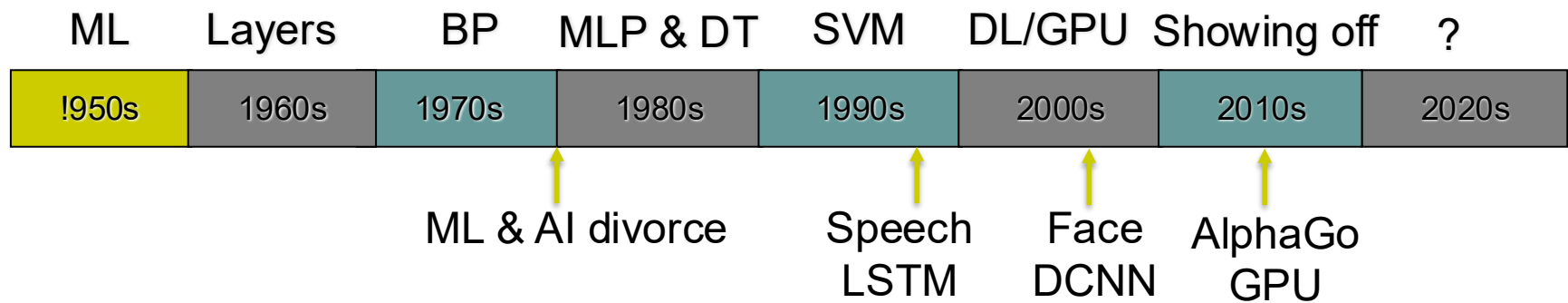
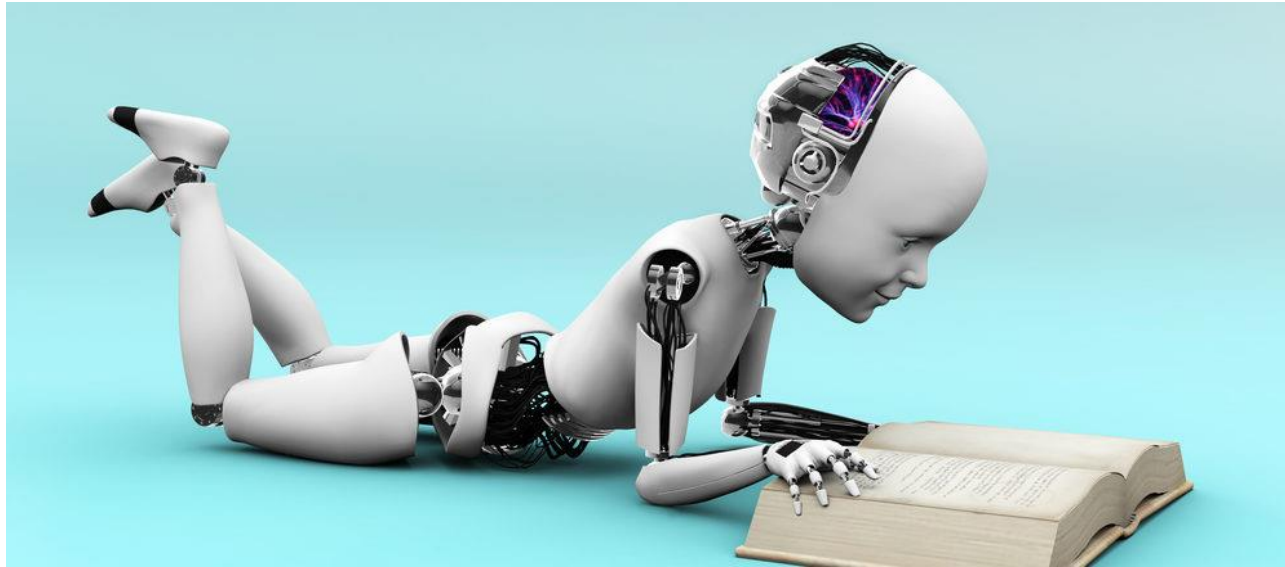
Abu Yūsuf Ya‘qūb al-Kindī (801–873)

- "Manuscript on Deciphering Cryptographic Messages"- relative frequency analysis
- Probability and statistics ([Pierre de Fermat](#) and [Blaise Pascal](#) in 1654)

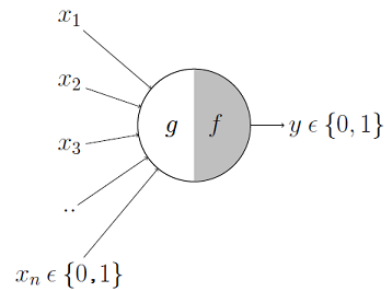




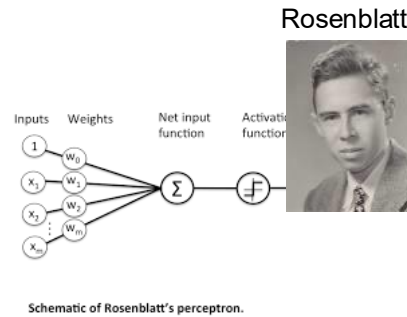
How it all evolved?



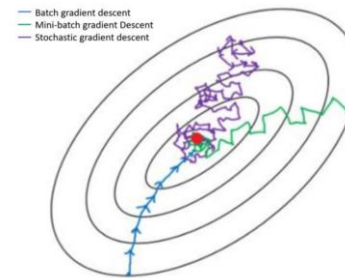
History of Deep Learning



40': McCulloch-Pitts Neuron



50': Rosenblatt perceptron (Father of DL)



60': ADALINE, Stochastic gradient descent



70': Backpropagation
1st Winter of AI

Linnainmaa



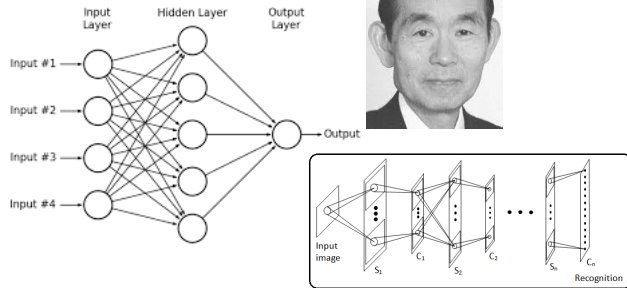
History of Deep Learning



Schmithuber

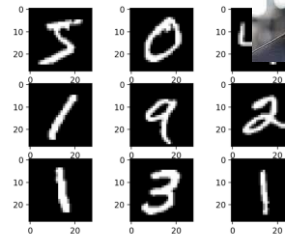


Fukushima

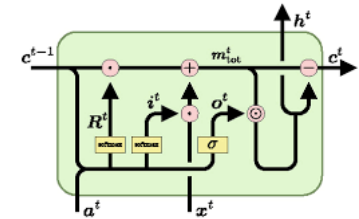


80': Multi-Layer Perceptron,
Neocognitron (predecessor of
CNN)

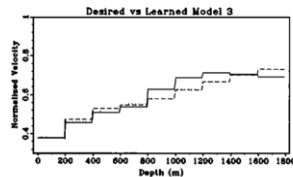
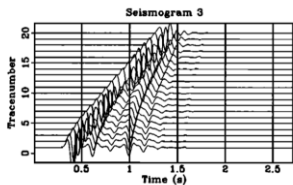
LeCun



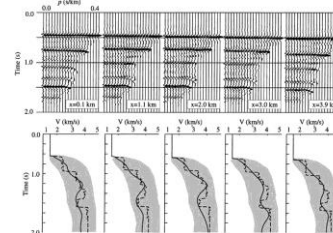
1989: LeNet



1997: Sequence
modelling and
LSTM



1994: Ruth and
Tarantola, NN
inversion of seismic
data



1999: Sen and Stoffa,
NN-based NMO
correction and
velocity estimation

History of Deep Learning

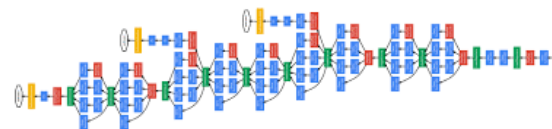


2000': **2nd Winter of AI**



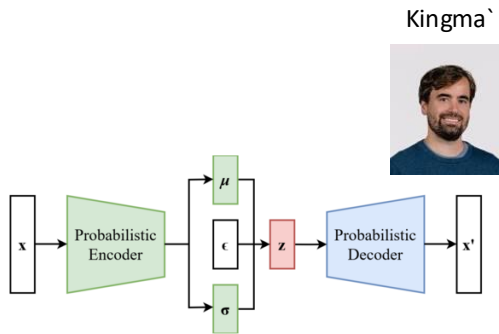
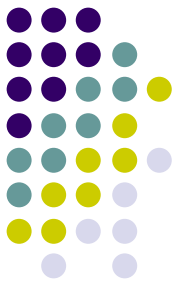
2012: ImageNet competition won by DeepCNN (AlexNet)

Krizhevsky

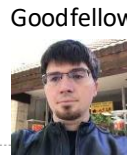
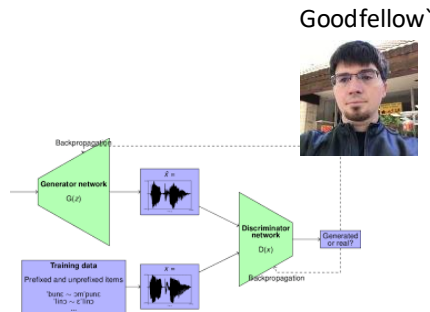


2010': **Explosion of DL**
New architectures (VGG, ResNet, GoogleLeNet...) and optimizers (RMSProp, ADAM...)

History of Deep Learning



2014: Variational AutoEncoders



2014: Generative Adversarial Networks (GANs)

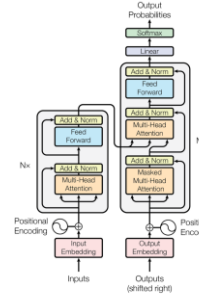


Figure 1: The Transformer - model architecture.

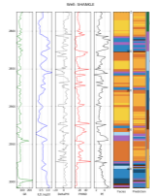
Bahdanau



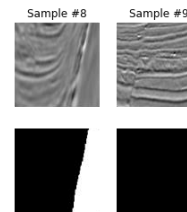
2015/7: Attention mechanism and Transformers

....

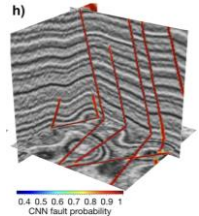
2016: SEG Challenge for facies classification



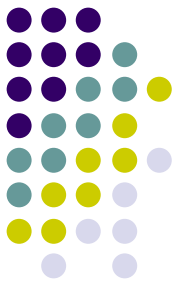
2019: TGS Kaggle Salt Interpretation Competition



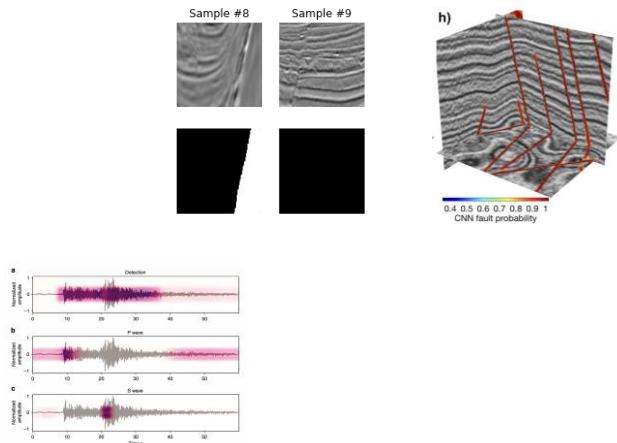
2019: FaultSEG h)



ML/DL Applications in Geoscience

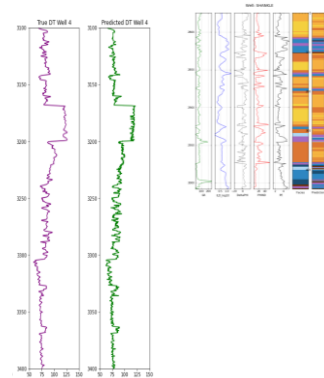


Seismic Interpretation (Faults, Horizons, Bodies)

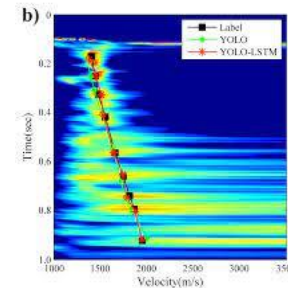


Passive seismic/Earthquake Seismology (Event picking, source localization and mechanism)

Well log Analysis (Facies Classification, Well log imputation...)

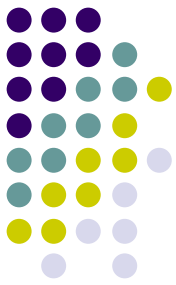


Interpretation tasks

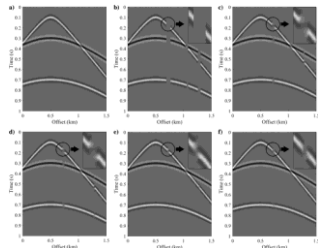
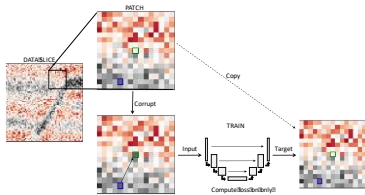


Seismic processing – manual tasks (Velocity picking, dispersion curve picking)

ML/DL Applications in Geoscience

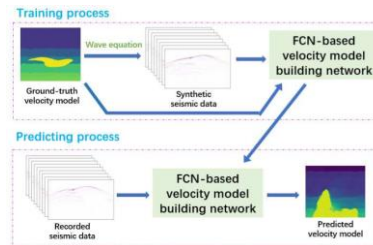


Seismic Denoising (Random, Ground Roll, Multiples)

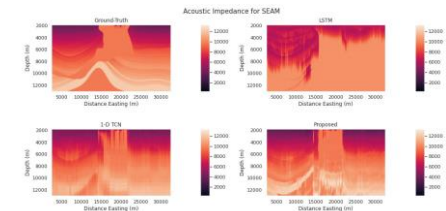


Seismic processing (Deghosting, Deblending, Interpolation, Acoustic-to-Elastic, 4D Matching, Low-frequency extrapolation...)

Velocity Analysis (CMP-based velocities, FWI, Salt flooding,)



Processing/imaging tasks



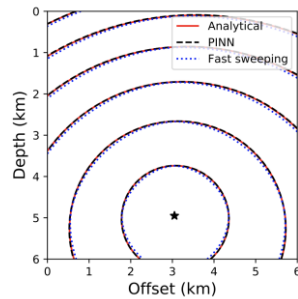
Seismic inversion (Post- and pre-stack inversion, rock physics inversion)

ML/DL Applications in Geoscience

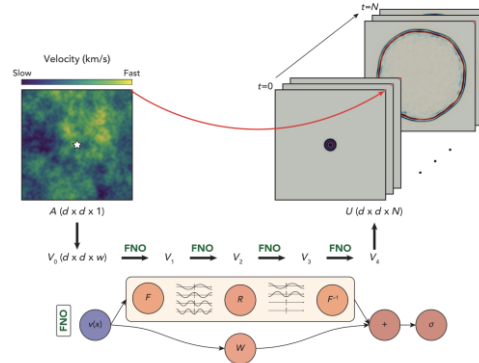


Modelling

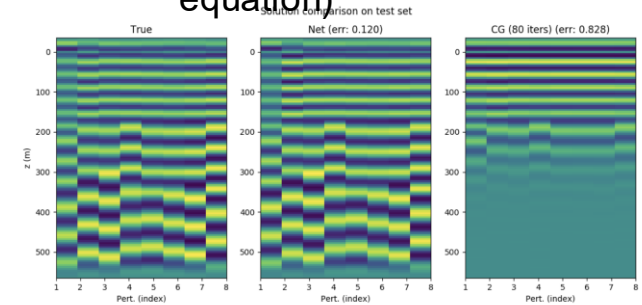
PINNs (Wave equation, Helmholtz,

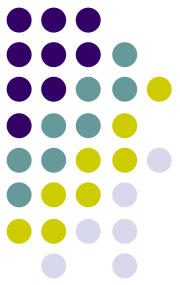


Fourier Neural Operators
(Wave equation)



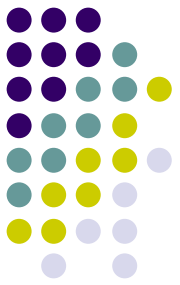
Learned Iterative solver (Helmoltz equation)





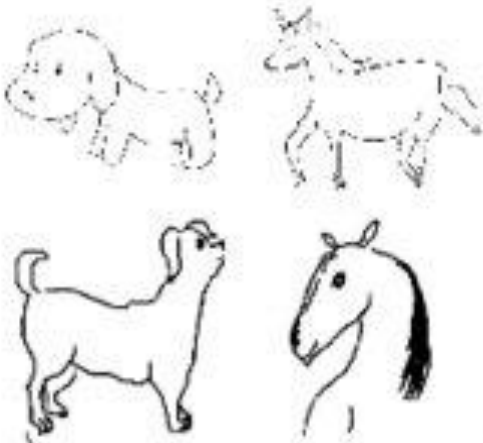
The decade of the Machine (2020s)

- Applications, Applications, Applications
- Connected cars that drive us to places
- A virtual assistant who understands us better than anyone
- We need knowledge of the domain of application
- Thus, ML in big data, image processing, text processing, sound,
- Geoscience applications
 - In Seismics: applications in image processing, horizon and fault detection, classification noise reduction, facies classification
 - Applications in deep physics/geoscience?



An ML challenge

Sketch



Cartoon



Art painting

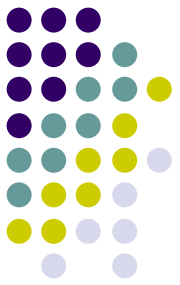


Photo



Training set

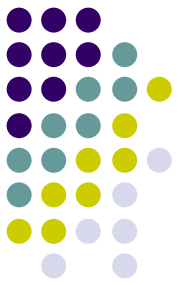
Test set



Lip reading? Is there something that ML can not do?



Should we Worry?



“Machine Intelligence is the last invention that humanity will ever need to make” — Nick Bostrom

“Success in creating AI would be the biggest event in human history. Unfortunately, it might also be the last, unless we learn how to avoid the risks” — Stephen Hawking

“AI doesn’t have to be evil to destroy humanity — if AI has a goal and humanity just happens to be in the way, it will destroy humanity as a matter of course without even thinking about it, no hard feelings.” — Elon Musk

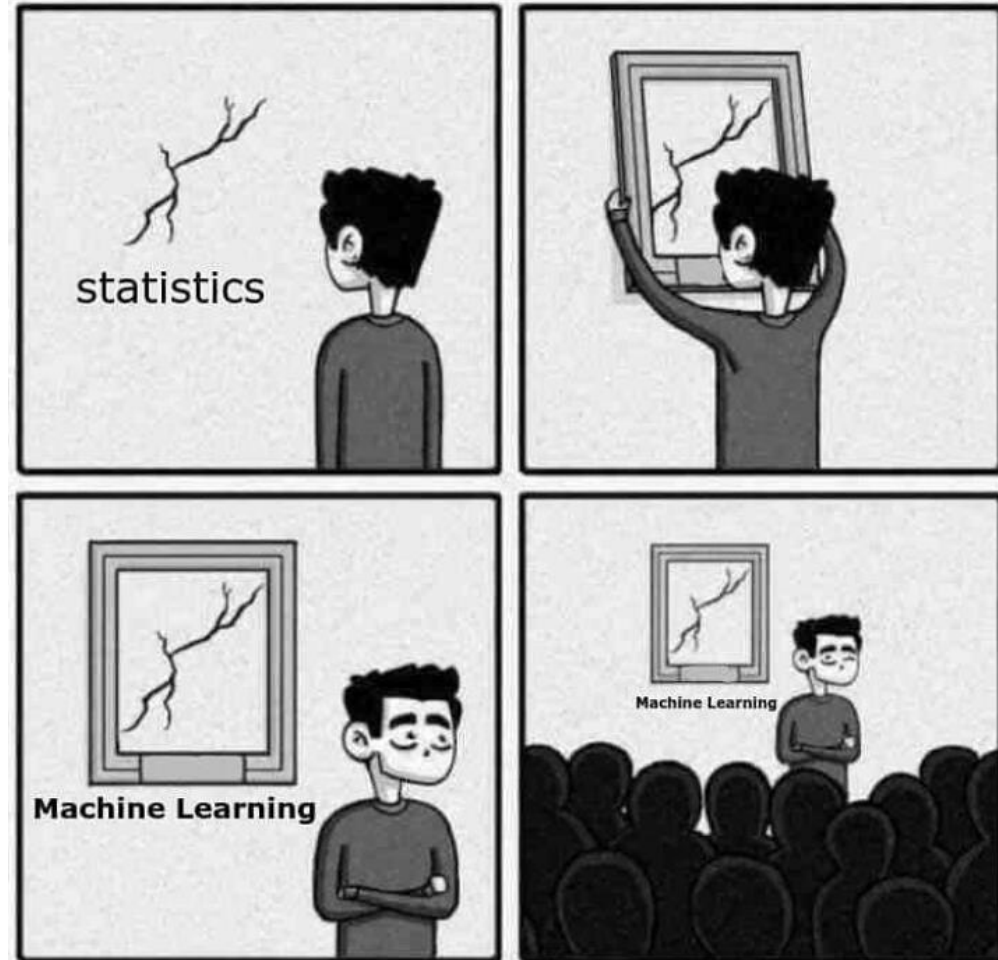




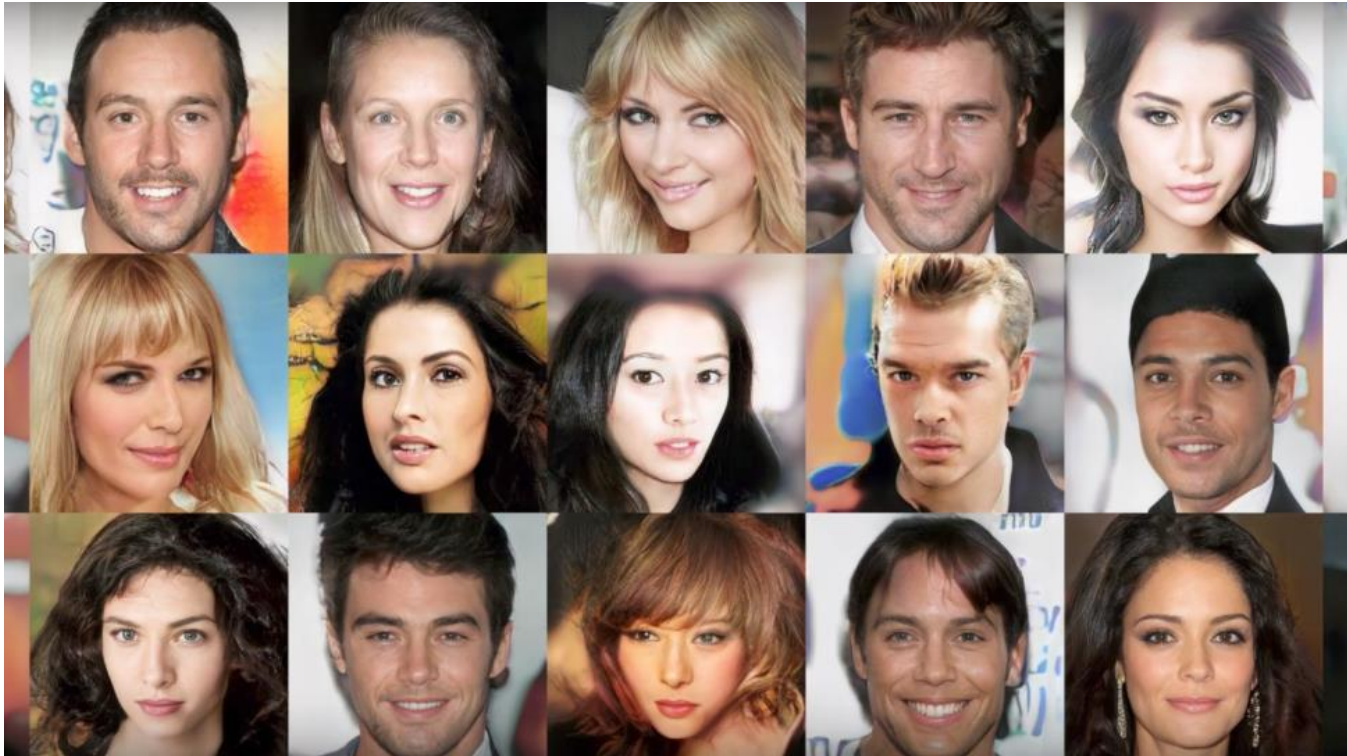
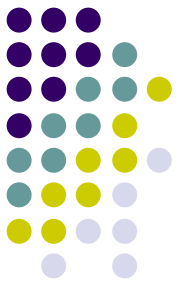
What is next?

- *Human intelligence 2029*
- *1000 fold human intelligence 2045*
- *Artificial General Intelligence*

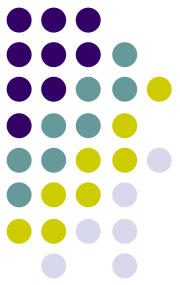
“As a technologist, I see how AI and the fourth industrial revolution will impact every aspect of people’s lives.” — Fei-Fei Li



Generative AI

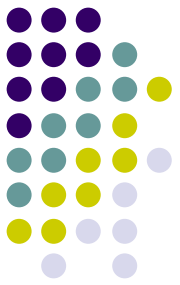


How?



Important terms in ML

- Model or hypothesis
- Model parameters, weights and biases
- Model architecture
- Training: pretraining, fine tuning and Pruning (optimization)
- Training data, validation, testing, labeled , labelless
- Type of training: supervised/unsupervised/self-supervised
- Overfitting/generalization
- Data bias
- Inductive bias
- Pruning, bagging, boosting,
- Regularization
- Bayesian NN
- Teacher, student, teacher forcing!



Some useful terminology of ML

- Supervised Learning:

$$X \rightarrow Y$$

$$x_i, y_i \rightarrow y_i = f_{\theta}(x_i)$$

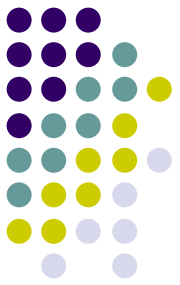
Regression
(Y continuous)

Classification
(Y discrete)

- Unsupervised Learning:

$X, X \rightarrow Z \rightarrow X$ Dimensionality Reduction

$P(X)$ Generative (learn PDF)



Some useful terminology of ML

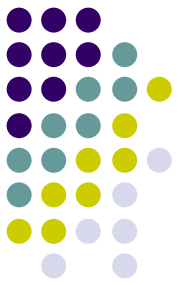
- Structured data: Tables, databases

In Geoscience: Well logs, production data, drilling data,...

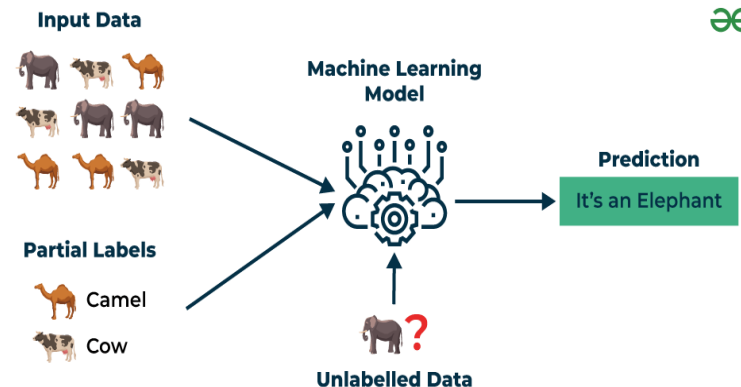
- Unstructured data: Images, audio, text

In Geoscience: Seismic data, Passive recordings, Gravity, EM, CT scans,...

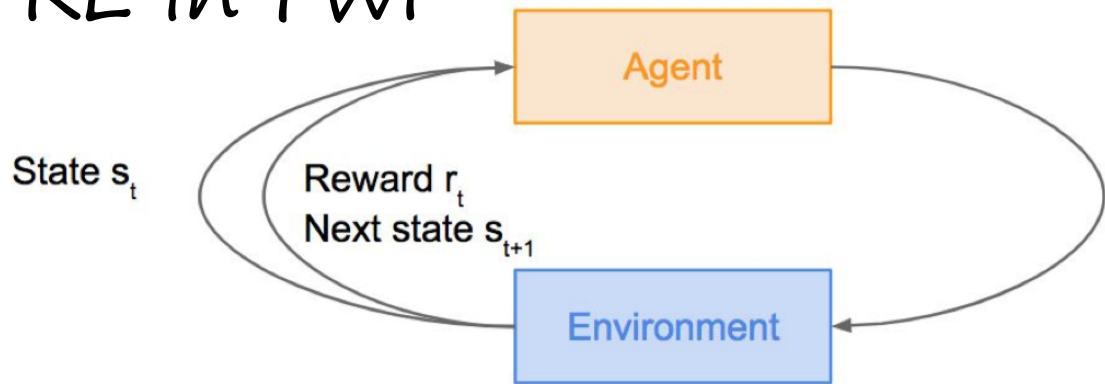
The kinds of training



- **Supervised:** labels/answers are used, learn to interpolate and possibly extrapolate between X and Y , depending on the kind of label (regression, classification).
- **Unsupervised:** no labels, learn to characterize the input , extract features in the data $\rightarrow$ clustering, dimensionality reduction.
- **Semi-Supervised:** A mix
- **Self-supervised:** where the labels are the input $X=Y$, maybe masking.
- **Reinforcement learning (RL):** using a reward system to guide the prediction.
Popular with self driving.
- **Data:** training, validation, testing.



RL in FWI

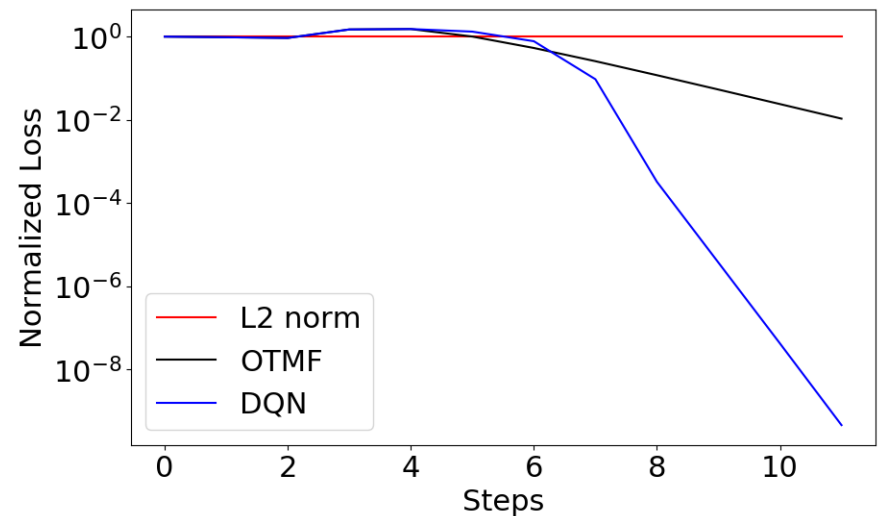
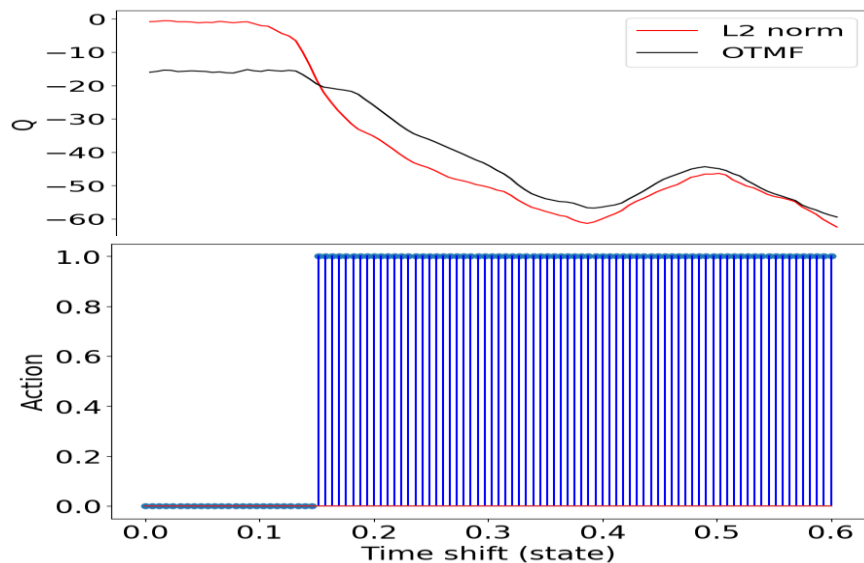
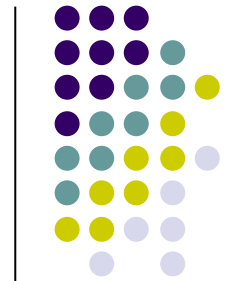


$$s_t = (p_t, d_t)$$

$$a_t = l_2 \quad \text{or}$$

$$\text{Action } a_t$$

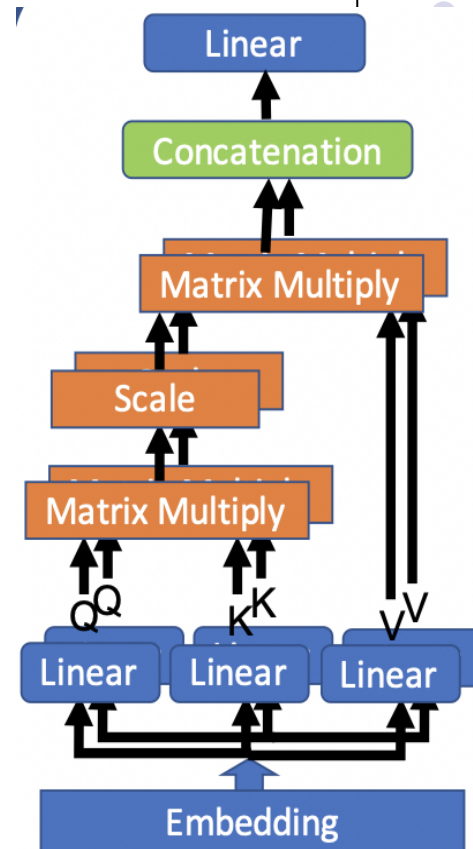
OTMF



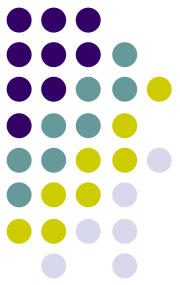
Sun, B., & Alkhalifah, T. (2020). A Data-Driven Choice of Misfit Function for FWI Using Reinforcement Learning. EAGE 2020 Annual Conference & Exhibition Online. doi:10.3997/2214-4609.202010203

Neural network operations

- **Feed forward/MLP:** Good in developing functions.
- **Convolutional:** Learned convolution filters $\rightarrow$ channels, local feature extraction.
- **Attention mechanism:** A special correlation-based weighted summation of vector sequences. More global.
- In addition to: pooling, skip connections, residual connections, pointwise multiplication, concatenation.....



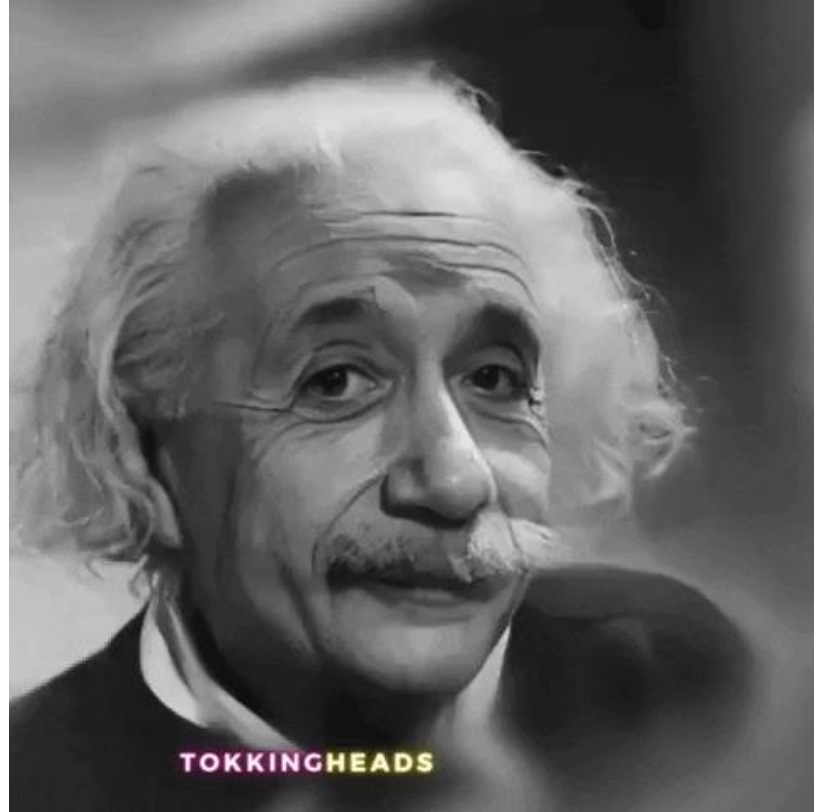
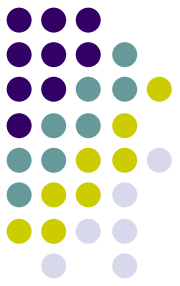
The attention mechanism



Neural network architectures

- **Encoder-Decoder:** AE, VAE, a famous kind Unet.
The key idea (the latent representation).
- **GANs (Decoder/Generator-Encoder/Discriminator):**
A generative model.
- **Recurrent NN (RNN):** Carries hidden state information from the previous prediction. Good for sequences.
- **Transformers:** Very popular with sequences, ChatGPT
- **Mixed**

Elbert Einstien





Thank You

30 17:50